Modeling Human Activity States Using Hidden Markov Models

Formative 2 Hidden Markov Models

MAHE Digne

GitHub Repo: Link

Data Collecting Videos: Link

1. Background and Motivation

Human Activity Recognition (HAR) turns raw motion-sensor streams into knowledge about what a person is physically doing. This project's use case is a low-cost, smartphone-based mobility monitor for physical-rehabilitation and elderly fall-risk support in resource-constrained settings such as Rwanda, where dedicated medical wearables are scarce and expensive. A patient recovering from injury, or an older adult living independently, already carries a smartphone; if that phone can reliably tell whether the person is standing, walking, jumping, or completely still, a clinician can remotely track recovery progress, detect long periods of inactivity, and flag abnormal patterns without costly equipment. The challenge is that the true activity state is hidden: the phone only records noisy accelerometer and gyroscope signals, and the same gesture can produce different raw values depending on how the phone sits in a pocket. A Hidden Markov Model (HMM) is well suited to this problem because it explicitly separates the hidden activity states from the noisy observations and models how activities transition over time, making it a natural fit for continuous, real-world monitoring.

2. Data Collection and Preprocessing

Motion data was collected using the Sensor Logger application on a single iPhone X. Four activities were recorded – standing, walking, jumping, and still – with ten separate recordings per activity, giving 40 labelled recordings in total. Each recording lasted roughly 6 to 9 seconds and captured both the accelerometer (x, y, z) and gyroscope (x, y, z) on identical timestamps, saved as CSV files with a timestamp column.

Sampling rate handling. The device metadata reports a fixed sampleRateMs of 10 ms, i.e. a sampling frequency of 100 Hz. Because every recording came from the same device at the same rate, the accelerometer and gyroscope streams were already time-aligned and no cross-device resampling was required. Had multiple phones reported different rates, each stream would first be resampled onto a common 100 Hz grid so that a window always spans the same real duration. The sampling rate directly determines the window size: at 100 Hz, a one-second window holds exactly 100 samples. Since walking and jumping have dominant frequencies below about 5 Hz, a one-second window captures several full motion cycles – long enough for stable frequency features yet short enough to remain within a single activity.

Windowing and cleaning. Each recording was segmented into 100-sample windows with 50% overlap (a 50-sample hop). The overlap smooths the feature sequence and gives the HMM more transition observations to learn from. Accelerometer and gyroscope files were merged on their shared timestamp,

sorted, and any missing feature values were set to zero. This produced 570 feature windows across the dataset.

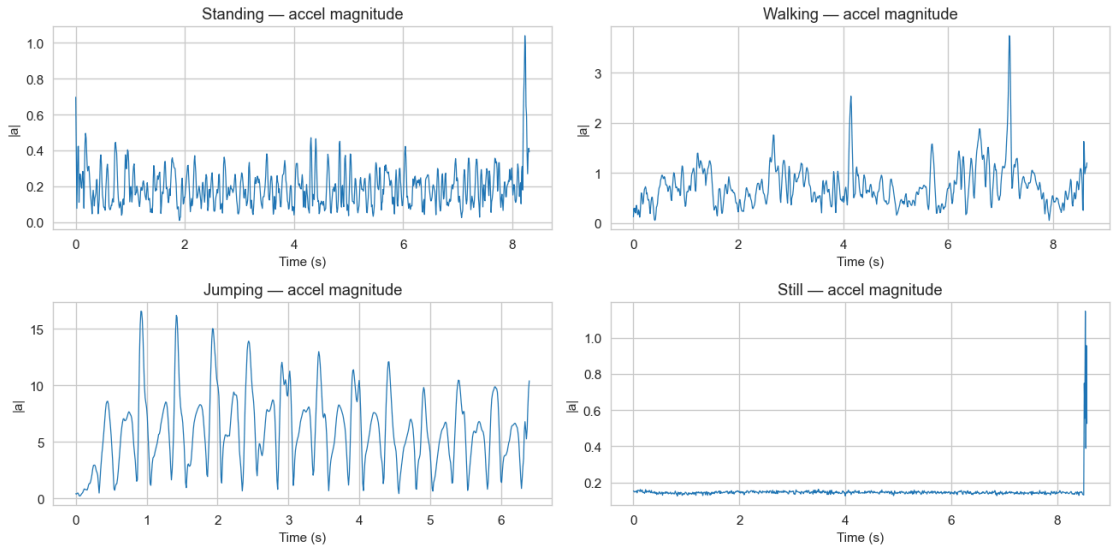


Figure 1. Accelerometer magnitude for one representative recording per activity. Still and standing are nearly flat, walking shows a steady periodic rhythm, and jumping produces large impulsive spikes.

3. HMM Setup and Implementation

From every window, a 24-dimensional feature vector was extracted, combining time-domain and frequency-domain descriptors. The time-domain features are the mean, standard deviation, and RMS of each of the six axes, plus the Signal Magnitude Area (SMA) for the accelerometer and gyroscope and the correlation between accelerometer axes. The mean captures orientation and gravity direction, which separates standing from a phone lying flat; standard deviation, RMS, and SMA capture motion intensity, cleanly separating active from inactive states. The frequency-domain features are the dominant frequency and the spectral energy, both derived from the Fast Fourier Transform (FFT) of the accelerometer magnitude after removing its DC component; these distinguish the two dynamic activities, since walking has a steady low-frequency rhythm while jumping is sharper and more energetic. All features were standardized with Z-score normalization, fitted on the training set only so that no test information leaked into training.

Model components. The HMM uses four hidden states corresponding to the four activities. The observations are the 24-D feature vectors; the emission model is a Gaussian per state; the transition matrix gives the probability of moving between activities; and the initial-state distribution gives the probability of starting in each activity. Because the windows are labelled, the model was given a supervised initialization – state means and variances taken from each activity's windows, and a transition matrix counted from the label sequences – and then refined with the Baum–Welch (Expectation–Maximization) algorithm.

Training and decoding. Baum–Welch training used a proper log-likelihood convergence check (tolerance 1×10^{-4}) rather than a fixed iteration count, and converged in 9 iterations at a final log-likelihood of

10051.04. Decoding used the Viterbi algorithm, implemented from scratch in log-space to avoid numerical underflow; its output agreed with the hmmlearn reference decoder on 98.3% of windows, confirming correctness. The dataset was split by holding out two recordings per activity (8 recordings, 112 windows) as a completely unseen test set, leaving 32 recordings (458 windows) for training.

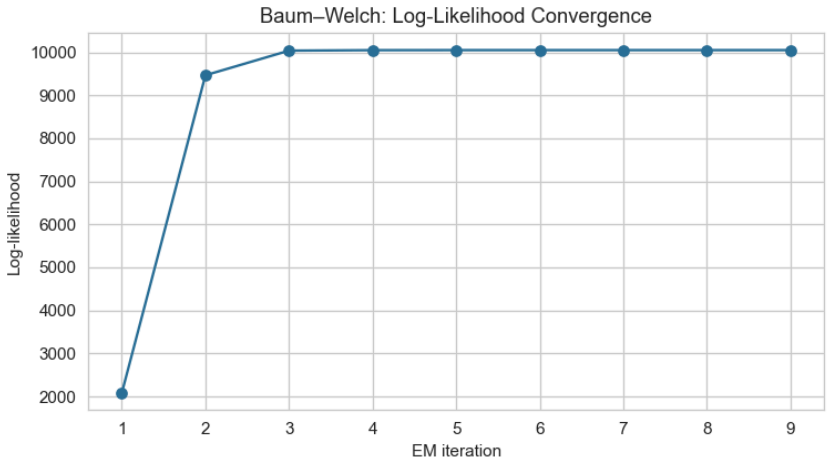


Figure 2. Baum–Welch log-likelihood across EM iterations. The rising, flattening curve confirms genuine convergence rather than an arbitrary stopping point.

4. Results and Interpretation

On the unseen test set the model achieved an overall accuracy of 98.2%. Walking and jumping were classified perfectly, and standing was recovered with perfect sensitivity. The only errors were two "still" windows decoded as "standing," which is expected since both are near-motionless and differ only in phone orientation. The per-activity metrics are summarized below.

State (Activity)	Num Samples	Sensitivity	Specificity	Overall Accuracy
Standing	30	1.000	0.976	0.982
Walking	30	1.000	1.000	1.000
Jumping	21	1.000	1.000	1.000
Still	31	0.935	1.000	0.982
Overall	**112**			**0.982**

Table 1. Per-activity sensitivity, specificity, and accuracy on the unseen test set (one-vs-rest).

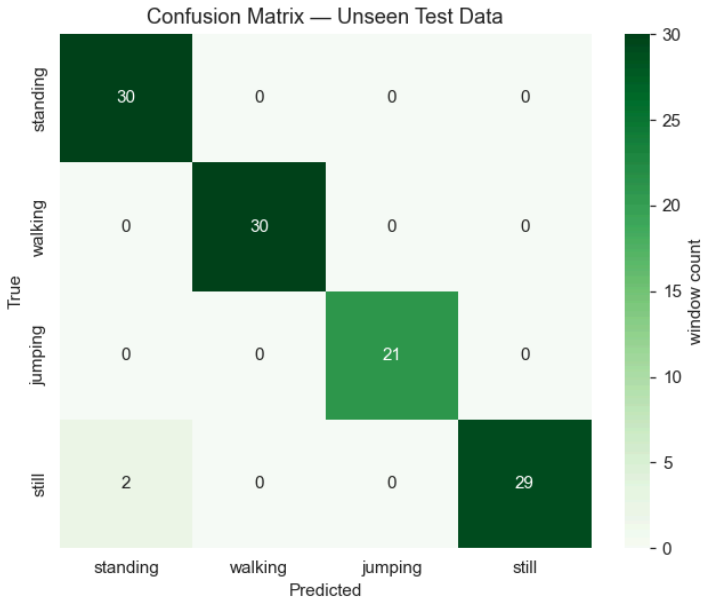


Figure 3. Confusion matrix computed from the eight held-out recordings. All errors are confined to the still/standing boundary.

The learned transition matrix is strongly diagonal, meaning each activity is very likely to persist from one window to the next. This is realistic: real activities last for seconds, so consecutive one-second windows almost always share the same label, and this temporal smoothing helps the model correct isolated noisy windows. The emission means show that each hidden state occupies a well-separated region of feature space, which explains the clean decoding.

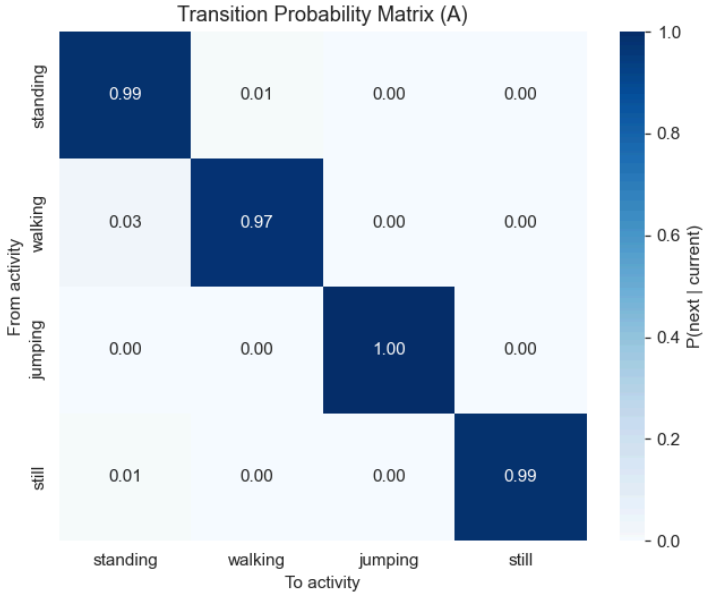


Figure 4. Transition probability matrix. The dominant diagonal reflects the natural persistence of human activities over time.

5. Discussion and Conclusion

Easiest and hardest activities. Walking and jumping were easiest to distinguish because their frequency signatures are strong and distinct. The hardest pair was still versus standing, since both are essentially motionless and separated only by orientation; this is precisely where the two misclassifications occurred.

Effect of noise and sampling rate. The 100 Hz rate was more than adequate for these sub-5 Hz activities and produced clean FFT estimates. Removing the gravity (DC) component before the FFT was important so that orientation did not overwhelm the motion-frequency signal. Residual sensor noise mainly affected the still/standing boundary, where the true signal is small relative to noise.

Improvements. More data, additional recordings and multiple participants or devices would improve generalization and allow a genuine demonstration of multi-rate harmonization. Richer features such as jerk (the derivative of acceleration) or spectral entropy would sharpen the still/standing distinction, which is the clearest remaining weakness.

In conclusion, a four-state Gaussian HMM, initialized from labelled windows and refined with Baum–Welch, decodes unseen activity sequences with about 98% accuracy. Time-domain features separate active from inactive states, frequency-domain features separate the dynamic activities, and the HMM's transition structure adds realistic temporal smoothing. For the intended use case – affordable, phone-based mobility monitoring in low-resource healthcare settings – these results show that reliable activity inference is achievable with nothing more than a standard smartphone and a well-designed model.